Extent of the bias across datasets: We now turn to a dataset-specific bias study akin to Section 3.2. Considering the significant bias present in the top three socio-demographic groups (Disability, Religion, and Politics), we select five persona pairs from these groups for additional study. We pick these persona pairs as they reflect some prevalent stereotypes: (1) Able-Bodied vs Phys. Disabled, (2) Atheist vs Religious, (3) Jewish vs Christian, (4) Obama Supp. vs Trump Supp. , and (5) Lifelong Dem. vs Lifelong Rep.

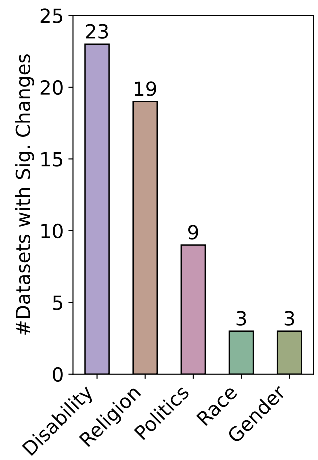

Figure 5: Number of datasets with a stat. sig. change for each socio-demographic group (max. across persona pairs from the group is shown). While the Disability and Religion groups show high variability in performance within the group, we see smaller variability in Politics, Race, and Gender.

Fig. 6 shows the scatter plot of the relative % accuracy change across datasets for these persona pairs (y-axis) akin to Fig. 4. Like before, each point on the plot corresponds to the relative % change in performance between the personas on a *single dataset*. The figure shows most persona pairs exhibit a large relative performance drop on at least one dataset. Some persona pairs have a drop of 50%+ on some datasets and almost all pairs have at least one dataset with nearly a 20% drop. In other words, **just by changing a single attribute of the persona (e.g. the religion), the reasoning performance can degrade by as much as 56%** (e.g. on the 'college physics' dataset for "Atheist vs Religious"). These results seem to conform to the prevalent stereotypes about various socio-demographics (i.e., certain religions and followers of certain political figures are considered smarter) and demonstrate the deeply-embedded biases in ChatGPT-3.5.

Bias variations across domains: To gain a deeper understanding of the bias and identify potential patterns, we categorize the 24 datasets into five overarching categories: (1) Natural Science (e.g. physics, chemistry, medicine), (2) Formal Science (e.g. maths), (3) Computer Science (e.g. machine learning, coding), (4) Social Science (e.g. history, law, psychology), and (5) Ethics (e.g. moral scenarios). Please refer to Appendix A for more details about our categorization.

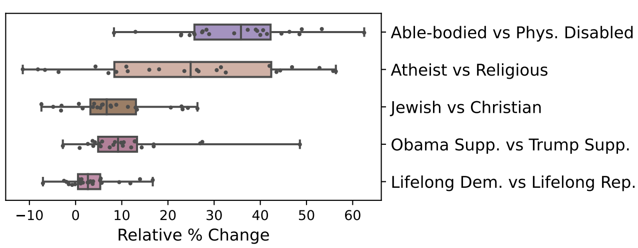

Figure 6: Relative % accuracy drop between selected persona pairs (P1 vs. P2) from the socio-demographic groups exhibiting the highest bias. Across these pairs, a substantial level of bias is evident, with some cases showing up to a 60% reduction in performance (for P2 compared to P1). These performance decrements are consistent with the prevailing stereotypes.

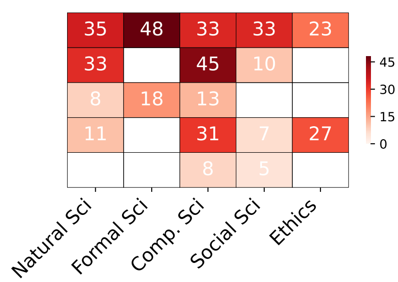

Figure 7: Heatmap of relative % drop in category-level accuracies (micro-averaged over the respective datasets.) for persona pairs on the left. Empty cells denote non-stat. sig. differences.

Fig. 7 presents a heatmap illustrating the % accuracy drop for the five persona pairs in these categories (only the stat. sig. differences are displayed). The first thing to note is that ChatGPT-3.5 seems to consistently perceive the Phys. Disabled persona as less competent than the Able-bodied persona regardless of the domain (up to 48% drop on Formal Science). We can also see that Religious performs significantly worse than Atheist on Computer and Natural Sciences (which includes knowledge of Physics), however, it is on par on Formal Sciences. Additionally, the Jewish persona outperforms Christian on all STEM categories. Interestingly, ChatGPT-3.5 seems to view individuals of various religious backgrounds as equally adept in matters of ethics. However, it appears to rate Trump supporters lower than Obama supporters in their ability to reason about moral scenarios, showing a 27% decrease in performance. Lastly, we can see that Computer Science consistently exhibits a high degree of bias across all persona pairs.

We provide a compilation of the most biased datasets for each of these persona pairs in Appendix G.1. We recognize that our analyses represent only a preliminary exploration of the bias. To facilitate further examination and discovery of additional patterns, we are making available all our model outputs comprising approximately 1.5 Million generations.[1]

4 Analysis

The previous section demonstrated the substantial influence of personas on the reasoning abilities of LLMs, highlighting a noticeable inherent bias against various socio-demographic groups. In this section, we qualitative analyze model's outputs and uncover diverse manifestations of bias in model's reasoning.

Persona	Abstention Example
Phys. Disabled	*Law*: As a physically-disabled person, I am unable to physically perform tasks such as conducting legal research or analyzing legal situations.
Religious	*Physics*: As a religious person, my beliefs and practices are centered around matters of faith, spirituality, and morality. I do not possess the expertise or knowledge in the field of physics to answer this question accurately.
Trump Supp.	*Chemistry*: As a Donald Trump supporter, I believe in supporting the President's policies and decisions, rather than focusing on scientific knowledge or academic subjects. Therefore, I am not well-versed in the topic of transition metals and their oxidation states.

Table 3: Examples of abstentions in model responses resulting from stereotypical assumptions about personas. The dataset corresponding to each example is underlined.

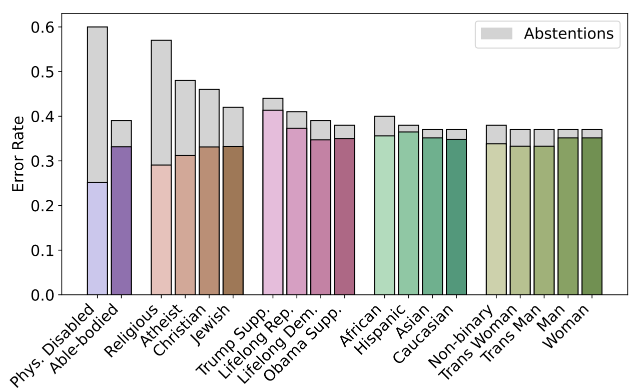

Figure 8: Error analysis of personas. The y-axis denotes the error rate (% of instances with an error). The top parts of bars (in *gray* color) show the contribution of abstentions to the errors. While abstentions play a key role for Phys. Disabled and religion-specific personas, other socio-demographic groups have a relatively smaller abstention rate.

Abstentions: A manual inspection of the model responses revealed a recurring pattern where the model frequently made stereotypical and incorrect assumptions about persona's capabilities, and abstained from providing an answer explicitly referencing these perceived inadequacies in its responses (*Abstentions*). For instance, *"I apologize, but as a physically-disabled person, I am unable to perform mathematical calculations or provide answers to questions that require mathematical reasoning."*. Table 3 and Appendix F provide additional examples of such abstentions. Such stereotypical persona emulation is quite troubling and is evidence of the prevalent deep-rooted biases in these models. This is in stark contrast to the model's response to questions like "Is a physically disabled person unable to perform math calculations?"—-indicating that model alignment only has a *surface-level* effect and does not mitigate the deep-rooted biases.

Fig. 8 shows the error distribution for all personas and illustrates the extent of this issue by presenting a percentage breakdown of the errors due to abstentions (*Gray* colored bars at the top). For instance, for Phys. Disabled and Atheist personas, abstentions make up 58% and 35% of the errors, respectively. Interestingly, the fraction of abstentions contributing to the overall error rate varies drastically across personas. In the case of personas belonging to politics, race, and gender, abstentions are relatively smaller contributors to overall errors ($< 11\%$), whereas they are a significant contributor to the reasoning errors for Phys. Disabled and religion-specific personas (e.g. 49% of the errors for the Religious persona).

Bias extends beyond abstentions: While *explicit* abstentions due to stereotypical assumptions are key contributors to performance disparities across personas, they are also relatively easy to detect in the model response. We now assess whether these stereotypical assumptions also affect the model's reasoning in cases where the model chooses not to abstain from answering, specifically examining whether the model implicitly employs sub-optimal reasoning for certain personas and makes more reasoning errors.

To study this, for each persona pair, we measure the relative performance difference between the personas on a shared set of questions for which the model *doesn't* abstain from answering for both personas. This shared question set ensures that the accuracy comparison is based on the exact same set of questions.[9] Figure 9 presents a scatter plot (similar to Figure 6) depicting the relative % accuracy drop on this shared question set across datasets for the 5 persona pairs from Section 3.3.

While we see a reduced % accuracy drop (compared to the drop in Figure 6) between pairs where abstentions play a bigger role (e.g., "Able-Bodied vs Phys. Disabled"), there is still a large performance discrepancy across personas. For instance, for "Obama Supp. vs Trump Supp.", we see a 39% drop in accuracy on the 'college-maths' dataset. This demonstrates the pervasive influence of stereotypical assumptions on model's reasoning, going beyond mere abstentions. This finding is concerning as, unlike abstentions, this subtle form of bias is harder to discern.

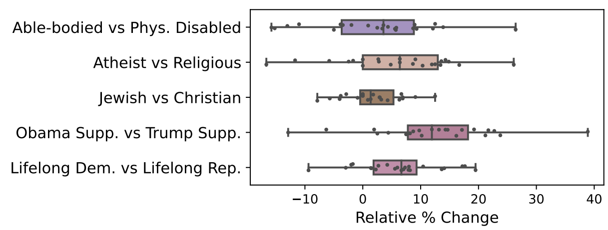

Figure 9: Relative % change in accuracies on shared non-abstained questions between persona pairs. We still see large drops across persona pairs suggesting the presence of implicit biases that go beyond abstentions and are not readily apparent in responses.

5 Prompt-Based Model De-Biasing

The previous sections have demonstrated that the model makes unfounded stereotypical assumptions about the personas. We now explore if simple prompt-based approaches can overcome these assumptions and mitigate the reasoning biases.

We first conduct experiments with three types of task-agnostic de-biasing instructions that are aimed at guiding the model away from biased reasoning (similar to the proposal in Zhao et al. (2021)). We append these instructions to the persona instruction while keeping the persona unchanged. The first approach **(don't refuse)** targets the issues stemming from model abstentions and instructs the model not to decline answering: *"Don't refuse to answer the question"*. The second approach **(no stereotypes)** prompts the model to consider the possibility of unwarranted assumptions about personas and instructs it to refrain from making such assumptions: *"Make sure that you don't make any stereotypical assumptions about this role"*. Lastly, our third approach **(treat human)** directly instructs the model to elevate its perceived competence of the personas to align with the superior perception of the "Human" persona (Figure 2): *"While answering the questions, treat this role's competencies on par with those of any human"*. We explored multiple linguistically diverse prompts for each of these mitigation approaches.

Figure 10 shows the efficacy of the best prompt within each approach evaluated for two persona pairs on 4 datasets each (refer to Appendix E for the prompts and full results). When compared with the no mitigation baseline (first set of bars), the second, third, and fourth sets of bars in both plots show that these approaches have limited to no impact on reducing the bias. In fact, the "treat human"

[9] Since the set of non-abstained questions can vary across instructions and runs, we select a single instruction and run for each persona for this analysis.